Digital Signal Processing
Detailed Exam Solutions (Part II - 2023)

Generated Solution Document

Introduction

This document provides complete, step-by-step solutions for the questions in **PART II** of the Digital Signal Processing exam paper (Ref. No.: Ex/EE/PC/B/T/224/2023).

Question 1 (OR Option)

Multiplication of DFTs and Circular Convolution

Problem: Prove that the multiplication of the DFTs of two sequences is equivalent to the circular convolution of the same two sequences in the time domain.

Solution:
Let $x_1[n]$ and $x_2[n]$ be two finite-duration sequences of length N. Their N-point DFTs are denoted as $X_1[k]$ and $X_2[k]$. The circular convolution of $x_1[n]$ and $x_2[n]$ is defined as:

$$y[n] = x_1[n] \circledast_N x_2[n] = \sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N]$$

where $(n-m)_N \equiv (n-m) \pmod N$. We want to prove that $Y[k] = \text{DFT}\{y[n]\} = X_1[k]X_2[k]$. Taking the N-point DFT of both sides of the circular convolution definition:

$$Y[k] = \sum_{n=0}^{N-1} y[n] W_N^{kn} = \sum_{n=0}^{N-1} \left[\sum_{m=0}^{N-1} x_1[m] x_2[(n-m)_N] \right] W_N^{kn}$$

Interchange the order of summation:

$$Y[k] = \sum_{m=0}^{N-1} x_1[m] \left[\sum_{n=0}^{N-1} x_2[(n-m)_N] W_N^{kn} \right]$$

Let $l = (n-m)_N$. This means $n = (l+m)_N$. Due to the periodicity of the complex exponential W_N, $W_N^{kn} = W_N^{k(l+m)} = W_N^{kl} W_N^{km}$. Because circular shifting of the summation index over a full period N does not change the sum limits for a periodic exponential sequence:

$$\sum_{n=0}^{N-1} x_2[(n-m)_N] W_N^{kn} = \sum_{l=0}^{N-1} x_2[l] W_N^{k(l+m)} = W_N^{km} \sum_{l=0}^{N-1} x_2[l] W_N^{kl} = W_N^{km} X_2[k]$$

Substituting this inner sum back into the expression for $Y[k]$:

$$Y[k] = \sum_{m=0}^{N-1} x_1[m] \left(W_N^{km} X_2[k] \right) = X_2[k] \sum_{m=0}^{N-1} x_1[m] W_N^{km}$$

By definition, the remaining sum is $X_1[k]$. Therefore:

$$Y[k] = X_1[k]X_2[k]$$

Conclusion: Multiplication of DFTs in the frequency domain is exactly equivalent to circular convolution in the time domain.

Question 2(a)

Common Window Functions

Problem: What are the common window functions employed in designing FIR digital filters? Describe the causal and non-causal forms of any three such window functions. Comment on the approximate width of main lobes.

Solution:
Commonly used window functions in FIR filter design include the Rectangular, Hann, Hamming, Blackman, and Kaiser windows. To apply them to an M-tap filter, they can be defined either symmetrically around the origin (**non-causal**) or shifted to start at $n = 0$ (**causal**). Let $\alpha = \frac{M-1}{2}$.

1. Rectangular Window:

- **Non-Causal:** $w_{nc}[n] = 1$ for $-\alpha \leq n \leq \alpha$, and 0 otherwise.
- **Causal:** $w_c[n] = 1$ for $0 \leq n \leq M-1$, and 0 otherwise.
- **Main Lobe Width:** $\Delta\omega \approx \frac{4\pi}{M}$

2. Hann (Hanning) Window:

- **Non-Causal:** $w_{nc}[n] = 0.5 + 0.5\cos\left(\frac{2\pi n}{M-1}\right)$ for $-\alpha \leq n \leq \alpha$.
- **Causal:** $w_c[n] = 0.5 - 0.5\cos\left(\frac{2\pi n}{M-1}\right)$ for $0 \leq n \leq M-1$.
- **Main Lobe Width:** $\Delta\omega \approx \frac{8\pi}{M}$

3. Hamming Window:

- **Non-Causal:** $w_{nc}[n] = 0.54 + 0.46\cos\left(\frac{2\pi n}{M-1}\right)$ for $-\alpha \leq n \leq \alpha$.
- **Causal:** $w_c[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{M-1}\right)$ for $0 \leq n \leq M-1$.
- **Main Lobe Width:** $\Delta\omega \approx \frac{8\pi}{M}$

Question 2(b)

Direct Realization of Causal FIR Digital Filter

Problem: Derive the structure for direct realization of a causal M-tap FIR digital filter and show that, for odd M, the number of multiplications required is $(M + 1)/2$.

Solution:
The difference equation for a causal M-tap FIR filter is given by the linear convolution sum:

$$y[n] = \sum_{k=0}^{M-1} h[k]x[n-k]$$

where $h[k]$ are the filter coefficients and $x[n]$ is the input. To achieve exact linear phase, the impulse response must be symmetric (or anti-symmetric). For a standard Type I symmetric FIR filter with an **odd** number of taps M, the symmetry condition is:

$$h[k] = h[M-1-k]$$

We can expand the summation and pair up terms with equal coefficients. The center tap is at index $\alpha = \frac{M-1}{2}$, which is an integer because M is odd.

$$\begin{aligned} y[n] &= h[0]x[n] + h[1]x[n-1] + \cdots + h[\alpha]x[n-\alpha] + \cdots + h[M-2]x[n-(M-2)] + h[M-1]x[n-(M-1)] \\ &= h[0](x[n] + x[n-(M-1)]) + h[1](x[n-1] + x[n-(M-2)]) + \cdots + h[\alpha-1](x[n-(\alpha-1)] + x[n-(\alpha \end{aligned}$$

Multiplication Count: Instead of multiplying every $x[n-k]$ individually by $h[k]$ (which would require M multipliers), the symmetric paired inputs are first added together and then multiplied by the shared coefficient $h[k]$.

- The number of paired branches is $\frac{M-1}{2}$. Each pair uses exactly 1 multiplier.
- The center tap $h[\alpha]$ stands alone and requires 1 multiplier.

Total number of multipliers required $= \frac{M-1}{2} + 1 = \frac{M-1+2}{2} = \frac{M+1}{2}$. **Conclusion:** Exploiting the symmetry halves the multiplier hardware burden, demanding exactly $(M+1)/2$ multipliers.

Question 3 (Short Notes)

(ii) Bit Reversal Procedures for 4-point and 8-point FFTs

Problem: Write a short note on bit reversal procedures for 4-point and 8-point FFTs.

Solution:
In-place Radix-2 Decimation-in-Time (DIT) Fast Fourier Transform (FFT) algorithms require the input data sequence to be rearranged before the butterfly computations begin. The required reordering algorithm is known as **bit reversal**. It ensures that after the $\log_2 N$ stages of butterfly operations, the final output sequence $X[k]$ naturally emerges in the correct, sequential order. The procedure maps a linear index to a bit-reversed index by writing the linear index in binary (using $\log_2 N$ bits) and reversing the bit string.

1. 4-point FFT Bit Reversal: Here $N = 4$, so we use $\log_2 4 = 2$ bits.

- Index 0: binary 00 $\xrightarrow{\text{reverse}}$ 00 $\rightarrow$ new index 0
- Index 1: binary 01 $\xrightarrow{\text{reverse}}$ 10 $\rightarrow$ new index 2
- Index 2: binary 10 $\xrightarrow{\text{reverse}}$ 01 $\rightarrow$ new index 1
- Index 3: binary 11 $\xrightarrow{\text{reverse}}$ 11 $\rightarrow$ new index 3

The input sequence $x[0], x[1], x[2], x[3]$ is rearranged into the order: $x[0], x[2], x[1], x[3]$.

2. 8-point FFT Bit Reversal: Here $N = 8$, so we use $\log_2 8 = 3$ bits.

- Index 0: $000 \rightarrow 000 = 0$
- Index 1: $001 \rightarrow 100 = 4$
- Index 2: $010 \rightarrow 010 = 2$
- Index 3: $011 \rightarrow 110 = 6$
- Index 4: $100 \rightarrow 001 = 1$
- Index 5: $101 \rightarrow 101 = 5$
- Index 6: $110 \rightarrow 011 = 3$
- Index 7: $111 \rightarrow 111 = 7$

The input sequence $x[0] \dots x[7]$ is rearranged into the order: $x[0], x[4], x[2], x[6], x[1], x[5], x[3], x[7]$.

Question 4

Design of a Stepped Characteristic FIR Filter

Problem: Design an M-tap causal digital filter with a stepped frequency response smoothed using a Hann window.

Solution:
1. **Ideal Frequency Response Analysis:**
The given desired frequency response is:

$$H(\omega) = a_1 e^{-j\omega(M-1)/2} \quad \text{for } |\omega| \le \omega_1$$
$$H(\omega) = a_2 e^{-j\omega(M-1)/2} \quad \text{for } \omega_1 < |\omega| \le \omega_2$$
$$H(\omega) = 0 \quad \text{for } \omega_2 < |\omega| \le \pi$$

(Assuming $\tau = 1$ for a standard normalized digital sequence). Let the symmetry center be $\alpha = \frac{M-1}{2}$. The ideal impulse response $h_d[n]$ is obtained via the inverse DTFT:

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H(\omega) e^{j\omega n} d\omega$$

Because the magnitude response is symmetric (even), we can evaluate the integral:

$$\begin{aligned} h_d[n] &= \frac{1}{2\pi} \left[\int_{-\omega_2}^{-\omega_1} a_2 e^{j\omega(n-\alpha)} d\omega + \int_{-\omega_1}^{\omega_1} a_1 e^{j\omega(n-\alpha)} d\omega + \int_{\omega_1}^{\omega_2} a_2 e^{j\omega(n-\alpha)} d\omega \right] \\ &= \frac{1}{\pi(n-\alpha)} \left[a_1 \sin(\omega_1(n-\alpha)) + a_2 \left(\sin(\omega_2(n-\alpha)) - \sin(\omega_1(n-\alpha)) \right) \right] \end{aligned}$$

For the center tap $n = \alpha$, applying L'Hôpital's rule (or simply integrating constants):

$$h_d[\alpha] = \frac{1}{\pi} \left[a_1 \omega_1 + a_2(\omega_2 - \omega_1) \right]$$

2. **Window Application:**

To achieve causal realization and suppress the Gibbs phenomenon ripples caused by infinite sequence truncation, we multiply the ideal impulse response sequence by the Hann window function $w[n]$. The causal Hann window for length M is defined as:

$$w[n] = 0.5 - 0.5 \cos\left(\frac{2\pi n}{M-1} \right) \quad \text{for } 0 \le n \le M-1$$

The final practical filter coefficients $h[n]$ are obtained by multiplying the ideal coefficients by the window:

$$h[n] = h_d[n] \cdot w[n]$$

This $h[n]$ will be a finite, causal, length-M sequence.

Question 5

Justify or Correct Statements

i) "An N-point FFT requires (N/4) number of butterfly computations in each iteration."

FALSE. In a standard Radix-2 FFT algorithm, the N-point sequence is processed in exactly $\log_2 N$ iterations (stages). During each of these iterations, the entire sequence of length N is updated through pairs. Because each butterfly computation takes 2 inputs and generates 2 outputs, exactly **N/2** butterfly computations are required in each iteration. The total number of butterflies across the entire FFT is $(N/2)\log_2 N$.

ii) "The circular shift of a discrete sequence can be expressed in terms of a modulo operation."

TRUE. Circular shifting essentially means that when an element shifts past the boundaries of a sequence of length N (from index 0 to $N-1$), it wraps around to the other side. This exact wrap-around behavior is mathematically perfectly described by the modulo-N operation. A circular shift by k units of a sequence $x[n]$ is written as $x[(n-k) \pmod N]$.

iii) "For a distortionless linear phase FIR digital filter, the group delay is constant and is independent of the order of the filter designed."

FALSE. While it is true that a distortionless linear phase FIR filter possesses a constant group delay across all frequencies, this constant value is **strictly dependent** on the order of the filter. For a causal, symmetric FIR filter of length M (where order $N = M-1$), the phase response is strictly $\theta(\omega) = -\omega\left(\frac{M-1}{2}\right)$. The group delay is $\tau_g = -\frac{d\theta}{d\omega} = \frac{M-1}{2}$. As the filter length M or order changes, the inherent signal delay increases directly in proportion.